

Multi-Omics Target Discovery in Parkinson's Disease

1. INTRODUCTION AND DISEASE CHOICE

I chose Parkinson's disease (PD) because it is the most common neurodegenerative movement disorder, pathologically characterized by the loss of dopaminergic neurons in the substantia nigra pars compacta and the presence of Lewy bodies containing aggregated α -synuclein (**SNCA**). I focused on substantia nigra tissue because it represents the primary anatomical site of disease onset and severe neuropathology, making its local transcriptional and proteomic profile directly relevant to the underlying disease process. This analysis answers the critical question: **Which genes show convergent evidence across independent genetic, RNA, and protein layers, and are therefore stronger target candidates than single-layer hits?** By integrating human genome-wide association studies (GWAS), transcriptomic expression, and mass spectrometry proteomics, this pipeline aims to distinguish true disease drivers from platform-specific noise or non-causal transcriptional changes.

2. DATA AND METHODS

2.1 Datasets

Three open-access datasets were integrated at the gene level. Because each layer comes from a different cohort with no shared samples, integration compares per-gene summary statistics rather than correlating layers within individual patients.

Layer	Dataset / Source	Samples / Size	Access
Genomics	GWAS Catalog, Parkinson disease (MONDO_0005180)	698 associations → 393 genes	Open
Transcriptomics	GSE7621, substantia nigra, Affymetrix GPL570 (GEO2R)	16 PD, 9 controls	Open
Proteomics	Licker et al. (2014), PRIDE PXD000427, substantia nigra	5 PD, 8 controls; 1,779 genes	Open

Table 1. Integrated data sources. A gated alternative with matched multi-omics (AMP-PD) was considered but requires a formal data-use application.

2.2 Processing & Data Harmonization

Genomics: Of 905 downloaded associations, 698 annotated as Parkinson disease were kept; 207 sub-trait associations (e.g., levodopa response, age at onset, disease progression, dementia in PD) were excluded because they measure disease course or treatment response rather than baseline risk. Multi-gene entries were split, and the strongest association (maximum $-\log_{10} p$) was kept per gene.

Transcriptomics: GEO2R output (54,318 probes) was reduced to 20,799 genes by removing 2,197 probes mapping to multiple genes and keeping the most significant probe per gene. Canonical dopaminergic markers (**TH**, **SLC6A3**, **DDC**) were positive in the raw output, indicating a control-vs-PD contrast; the sign was flipped so all values represent PD vs control.

Proteomics: Per-protein PD/control ratios were extracted from the supplementary report (1,783 of 1,795 proteins recovered). The supplement provides a significance flag but no numeric p -values; direction was confirmed using proteins validated in the original paper (ferritin-L and nebullette increased in PD).

2.3 Integration and Scoring Scheme

Tables were joined on uppercase gene symbols. RNA and protein shared 1,684 genes, and 34 genes were present in all three layers; an inner join was executed across all three modalities. Concordance was defined as directional agreement in RNA and protein log-fold changes. Each layer's evidence ($|\log_2 FC|$ for RNA and protein, $-\log_{10} p$ for GWAS) was min-max normalized and combined with equal weights (1/3 each):



3. RESULTS: TOP TARGETS AND KNOWN BIOLOGY

Rank	Gene	RNA log ₂ FC	Protein log ₂ FC	Protein Sig.	GWAS -log ₁₀ p	Concordant	Score	Open Targets PD Score
1	LMNA	3.65	0.16	Yes	20.7	Yes	0.517	0.095
2	PBXIP1	-1.59	0.20	Yes	38.7	No	0.400	none
3	SNCA	0.29	-0.04	No	169.4	No	0.398	0.871
4	MAP4K4	-0.33	0.36	Yes	12.1	No	0.374	0.352
5	WDR1	-0.56	0.32	Yes	6.1	No	0.352	none
6	KTN1	-1.11	-0.20	No	9.7	Yes	0.296	0.150
7	BAG3	-0.04	0.30	Yes	11.7	No	0.293	0.411
8	NDUFAF2	0.74	-0.20	No	14.3	No	0.271	0.368
9	MAPT	0.72	0.06	No	68.0	Yes	0.243	0.678
10	PAM	0.42	-0.20	No	8.7	No	0.230	0.015
11	SH3GL2	1.22	-0.07	No	20.1	No	0.209	0.542
12	PPP6R2	-0.90	-0.12	No	7.2	Yes	0.196	0.013
13	CHL1	-0.75	0.14	No	5.4	No	0.195	0.013
14	FYN	-1.26	0.06	No	14.4	No	0.184	0.452
15	STK39	0.35	0.08	No	41.4	Yes	0.181	0.543

Table 2. Top 15 integrated candidates ranked by composite multi-evidence score. Open Targets PD score indicates canonical target-disease association (MONDO_0005180).

3.1 Recovery of Canonical PD Biology & Missing Hits

The integrative multi-omics pipeline successfully recovered well-known Parkinson's disease loci, with 13 of the top 15 candidates possessing prior Open Targets association scores. High-scoring canonical targets include **SNCA** (0.871), **MAPT** (0.678), **STK39** (0.543), **SH3GL2** (0.542), **FYN** (0.452), and **BAG3** (0.411). The top ranks of **SNCA** (#3) and **MAPT** (#9) were driven almost exclusively by extraordinary human genetic signals ($-\log_{10} p$ of 169.4 and 68.0, respectively) rather than prominent RNA or protein fold changes in post-mortem tissue. This indicates that genetic variation at these loci confers baseline risk, whereas bulk transcriptomic and proteomic alterations at late disease stages may be subtle or localized.

Conversely, major PD targets such as **LRRK2** and **GBA1** were notably absent from the top 15 table. This exclusion occurred because neither protein was detected in the Licker et al. proteomic dataset, which captured only ~1,800 abundant proteins. Because the pipeline enforced a strict three-way inner join, low-abundance signaling kinases (**LRRK2**) and membrane-bound enzymes (**GBA1**) were dropped regardless of their genetic importance.

3.2 Analysis of Non-Obvious Target Candidates

PBXIP1 (#2) and **WDR1** (#5) emerge as strong non-obvious targets without existing Open Targets associations with PD, yet both possess solid genetic support ($-\log_{10} p$ of 38.7 and 6.1) and statistically significant proteomic shifts. Most striking is **LMNA** (#1), which ranks first overall due to massive transcriptomic elevation ($\log_2 FC$ 3.65, $p = 0.0013$), significant protein upregulation ($\log_2 FC$ 0.16), and strong genetic linkage ($-\log_{10} p = 20.7$). Despite these top-tier metrics, Open Targets assigns **LMNA** a low association score (0.095) derived mainly from general literature rather than direct genetics. This discrepancy underscores the difference between mapped genes and causal drivers: nearest-gene GWAS assignment may link variants to structural nuclear proteins like **LMNA**, but its substantial expression spike in substantia nigra suggests it may reflect cellular stress, nuclear membrane perturbation, or accelerated neuronal aging triggered by disease.

4. DISCORDANCE BETWEEN RNA AND PROTEIN

This cross-layer discordance is not a computational error; rather, it highlights critical biological regulation in neurodegenerative tissue:

- **Protein Stability & Slow Turnover:** Decreased transcript synthesis can be offset by prolonged protein half-lives or impaired proteasomal clearance in degenerating neurons.
- **Post-Transcriptional Regulation:** Translational repression or microRNA targeting can uncouple mRNA transcript abundance from steady-state protein levels.
- **Pathological Aggregation:** In PD substantia nigra, misfolded proteins aggregate into insoluble inclusions, preventing degradation and artificially inflating measured protein levels relative to decaying mRNA.
- **Cohort and Cell-Type Heterogeneity:** Because datasets derived from unmatched cohorts, variation in post-mortem interval (PMI), disease stage, or cell-type composition (e.g., loss of dopaminergic neurons vs. reactive astrogliosis) across groups can create apparent cross-layer divergence.

5. SCOPE OF CLAIMS AND STUDY LIMITATIONS

Valid Claims: We can confidently claim a prioritized list of candidate targets supported by convergent, multi-layered evidence across human genetics, transcriptomics, and proteomics. These candidates serve as an unbiased starting point for downstream druggability and structural pocket evaluations.

Invalid Claims: We cannot claim direct causality, as post-mortem tissues reflect late-stage disease states. We cannot infer patient-level RNA-protein correlation because the datasets were generated from separate, unmatched cohorts. Furthermore, GWAS-mapped genes cannot be definitively claimed as causal drivers without functional fine-mapping.

Key Methodological Limitations

- **Cell-Loss Confound:** Massive dopaminergic neuron loss in late-stage PD brains means downregulated genes in bulk tissue may reflect neuron depletion rather than cellular transcriptional repression.
- **Small Cohort Sizes & Shallow Proteomics:** The proteomic layer evaluated only 5 PD vs. 8 control brains and was restricted to ~1,800 abundant proteins without numeric *p*-values.
- **Strict Join Bottleneck:** Enforcing a three-way intersection reduced the working candidate pool from thousands down to just 34 surviving genes.
- **Min-Max Compression:** *SNCA*'s massive GWAS score (169.4) compressed the relative genetic contribution of all other targets during normalization.

6. CONCLUSION AND WEEK 3 CARRY-FORWARD CANDIDATES

By integrating human genetics, transcriptomics, and proteomics, this study identified 34 tri-layer convergent candidates in the Parkinson's disease substantia nigra. The top 3 candidates recommended for Week 3 druggability assessment are:

1. **LMNA (Rank #1):** Top overall multi-evidence score driven by massive transcriptomic elevation ($\log_2FC$ 3.65) and significant protein increase, pointing to nuclear envelope stress.
2. **MAP4K4 (Rank #4):** High multi-layer evidence score with significant protein elevation and established target-disease association (0.352), offering a tractable kinase target.
3. **BAG3 (Rank #7):** Significant protein elevation combined with strong genetic support and high Open Targets score (0.411), representing a key regulator of autophagy and protein aggregation.

REFERENCES

- Barrett, T., Wilhite, S. E., Ledoux, P., Evangelista, C., Kim, I. F., Tomashevsky, M., ... Soboleva, A. (2013). NCBI GEO: Archive for functional genomics data sets—update. *Nucleic Acids Research*, 41(D1), D991–D995.
- Lesnick, T. G., Papapetropoulos, S., Mash, D. C., Ffrench-Mullen, J., Shehadeh, L., de Andrade, M., ... Maraganore, D. M. (2007). A genomic pathway approach to a complex disease: Axon guidance and Parkinson disease. *PLoS Genetics*, 3(6), e98.
- Licker, V., Turck, N., Kövari, E., Burkhardt, K., Côte, M., Surini-Demiri, M., Lobrinus, J. A., Sanchez, J.-C., & Burkhardt, P. R. (2014). Proteomic analysis of human substantia nigra identifies novel candidates involved in Parkinson's disease pathogenesis. *Proteomics*, 14(6), 784–794. <https://doi.org/10.1002/pmic.201300342>

